

Chapter 2

Uninflectability: Concepts and consequences

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Uninflectable lexemes are to be distinguished from simply uninflecting lexemes (e.g. English prepositions) or defective lexemes. Uninflectability can be an inherent lexical property or a contextually/constructionally defined phenomenon exhibited by otherwise normally inflecting lexemes. Uninflectability can also be total (Russian nouns such as *kenguru* ‘kangaroo’) or partial (Polish *muzeum*-nouns, Nakh-Dagestanian “sporadic agreement”). This implies a four-way typology, of which three types are attested in the literature. I argue that the missing type, contextual partial uninflectability, is instantiated by infinitive phrases in Hungarian and genitive plural based compounds in Latvian. I explore which types of inflectional processes are subject to uninflectability, providing evidence that it applies solely to the “core” synthetic inflection of a language, and not, say, to affix-clitics, periphrasis, evaluative morphology. I sketch a formal account of uninflectability within the framework of Paradigm Function Morphology, Version 2 (Stump 2016): an uninflectable lexeme is associated with the expected complete content paradigm for lexemes of its class, but lacks the form/realized paradigms, in whole or in part. I define lexical insertion as the insertion of the most highly specified value of the (multivalued) FORM attribute in the lexical representation (following Spencer’s 2013 model of lexical relatedness). Uninflecting lexemes such as English prepositions lack both content and form/realized paradigms and so lexical insertion is defined over the (sole) value of FORM, the lexeme’s root. In effect, an uninflectable lexeme likewise defaults to the root form, since this is the only form available in the absence of a form/realized paradigm. This account raises open questions about the architecture of morphology, for instance, the nature of ‘core’ synthetic inflection as opposed to other types of lexical relatedness, and the way that defectiveness should be treated in a formal model.



1 Introduction

In inflecting languages, some (classes of) lexemes are *uninflecting*, i.e. are not associated with any inflectional paradigm: *of, the, almost, ...*, but some members of otherwise inflecting classes may (unexpectedly) fail to inflect, i.e. they show (*lexical*) *uninflectability*. Spencer (2020) outlines three types of uninflectability, distinguishing lexical from constructional/contextual uninflectability, and total uninflectability from partial uninflectability. In this paper I argue that the two parameters of uninflectability define a four-way typology. Cases of total and partial lexical uninflectability are well-known, and it is not difficult to identify cases of total constructional/contextual uninflectability. In Section 2 I suggest that Hungarian and Latvian may provide instances of the fourth type, thus establishing the phenomenon of uninflectability as typologically robust.

An account of uninflectability owes an account of what it means for a lexeme to be inflected in the first place. In Section 3 I cite data suggesting that otherwise uninflectable lexemes nonetheless permit affix-like clitics, periphrastic exponence and evaluative morphology (as well as straightforward derivational morphology). I propose that what the uninflectable lexeme is unable to inflect for is some kind of “core” synthetic inflectional morphology, closer to the canonical ideal (see Corbett, this volume, on Canonical Typology).

In Section 4 I propose a formal account based on the distinction drawn in Paradigm Function Morphology, Version 2 (PFM2) (Stump 2016) between the content paradigm on the one hand, and the form/realized paradigms on the other. The leading idea is very simple. An uninflecting lexeme, such as an English preposition, has just a (root) form and neither content nor form paradigm. An uninflectable lexeme, on the other hand, has a content paradigm and therefore behaves as though it were an inflecting lexeme, but it lacks a form/realized paradigm. Uninflectability is not the same as defectivity: it is not the case that an uninflectable lexeme cannot be used in sentences. But this raises the question of what form of an uninflectable lexeme it is that undergoes lexical insertion. I propose a solution to this problem which appeals to an enriched notion of lexical representation, based on the model presented in Spencer (2013).

The discussion illustrates an important point made by one of the first linguists to investigate uninflectability (in Slavic languages) on a large scale: “Unflektierbarkeit zeigt die Grenzen und damit die innere Struktur des morphologischen Systems einer Sprache” (Doleschal 2002: 60; see also Doleschal 2001): “Uninflectability points to the boundaries and hence the inner structure of the morphological system of a language”. I would argue, though, that the importance of uninflectability goes beyond the implications for any given language, but has reper-

cussions for the way we view morphological systems in general, and provides further evidence in support of an autonomous morphology (cf Aronoff 1994), and in support of the idea that inflectional systems are structured in paradigms.

2 A typology of uninflectability

In Spencer (2020) I argue for two axes or parameters of uninflectability. The familiar instances are those where a single lexeme fails to inflect in all morphosyntactic contexts, so that uninflectability is an idiosyncratic lexical property of that particular lexeme. This is *lexical uninflectability* (corresponding to Doleschal’s (2001) ‘paradigmatische Uninflektierbarkeit’). However, there are also cases in which an otherwise inflecting class of lexemes fails to inflect in a specific morphosyntactic context or construction. This is *contextual* or *constructional uninflectability* (Doleschal’s 2001 “systematische Uninflektierbarkeit”). For instance, the non-head noun of an English noun-noun compound typically fails to take (plural) inflection. The second parameter is that of total vs. partial uninflectability. In the most familiar cases an uninflectable lexeme has no inflected forms at all – *total uninflectability*. However, there are well-documented instances in which a lexeme inflects for one set of features, say, for nominal case in singular number forms, but fails to show inflection for other feature combinations, e.g. nominal case in the plural. This is *partial uninflectability*. The hypothetical example just give in fact corresponds to the Polish *muzeum*-nouns discussed below, p. 61.

This classification of uninflectability in terms of total vs. partial and lexical vs. constructional/contextual uninflectability implies a four-way typology, shown schematically in Table 1: Type 1 total/lexical, Type 2 partial/lexical, Type 3 total/contextual, Type 4 partial/contextual. All but Type 4 have been discussed in the literature, but before I come to Type 4 I very briefly exemplify the first three types.

Table 1: Four theoretical types of uninflectability

	lexical	contextual
total	Type 1	Type 3
partial	Type 2	Type 4

A clear case of Type 1, total lexical uninflectability, is found in the Russian noun lexicon. Russian nouns distinguish $6 \times 2 = 12$ case/number forms but about

3000 are indeclinable (uninflectable): all the forms of their paradigm are identical to the root, e.g. *kenguru* ‘kangaroo’. We distinguish such uninflectable lexemes from uninflecting lexemes on the one hand and defective lexemes (e.g. Russian *mečta* ‘dream’, lacking genitive plural) on the other. Uninflectable lexemes can occur in all the contexts open to inflecting lexemes (unlike defective lexemes), but the form is invariable, as shown in the Russian examples in (1) (where *uninfl* indicates an uninflecting or uninflectable form):

- (1) a. *odin vombat/kenguru*
 one.NOM.M.SG wombat(M)NOM.SG/kangaroo(UNINFL)
 ‘one wombat/kangaroo’
 b. *s èt-imi vombat-ami/kenguru*
 with these-INS.PL wombat-INS.PL/kangaroo(UNINFL)
 ‘with these wombats/kangaroos’

Crosslinguistically, any part of speech, not just nouns, can show uninflectability.

In a number of languages, certain classes of adjective borrowed from languages with rather different inflectional systems fail to inflect: German *prima*, ‘great’, *rosa*, ‘pink’, Greek *ble* ‘blue’, *gri* ‘grey’. In Bulgarian and Macedonian a number of adjectives borrowed from Turkish are uninflectable, such as Macedonian *taze* ‘fresh’ (Friedman 1993. See also Thornton & D’Achille, this volume, for discussion of uninflectable adjectives in Italian.)

Some languages have inflecting adpositions. These are often derived historically from nouns, and they typically take possessive agreement morphology. For instance, Welsh prepositions are said to fall into three inflectional classes: AR ‘on’, *arnaf* ‘1SG’, *arno* ‘3SG.M’, *arni* ‘3SG.F’ etc, TROS ‘over’, *trosof* ‘1SG’, *trosto* ‘3SG.M’, *trosti* ‘3SG.F’ etc, GAN ‘with’, *gennyf* ‘1SG’, *ganddo* ‘3SG.M’, *ganddi* ‘3SG.F’ etc. However, many prepositions fail to inflect, and instead just take ordinary pronominal complements (Williams 1980: 47, 127–130): *GYDA* ‘with’ *gyda mi* ‘with me’.

While less commonly found, even verbs can be uninflecting, as in the case of French loans from Romani, verbs in the Verlan criminal argot, and more recently loans from English such as *follow* ‘follow (e.g. on the social media platform X, formerly known as “Twitter”)’ (Boyé 2019). Examples of the latter can be found in social media and in the lyrics of French rap songs, as in (2). (These examples are of interest for the way we will later come to characterize the notion of *inflection*.)

- (2) a. Baby girl, si je te follow.
 baby girl, yes I you follow
 ‘Yes, babe, I’m following you.’ (Sirène – Luidji)
<https://music.youtube.com/watch?v=0EeHeHnqQio>

- b. Elle m'-a follow.
 she me-has follow
 'She (has) followed me.'
<https://www.youtube.com/watch?v=GbzBfr2aBus>
- c. Elle m'-a pas followback quand je l'-ai follow.
 she me-has NEG followback when I her-have follow
 'She didn't follow me back when I followed her.'
<https://www.azlyrics.com/lyrics/niska/rseaux.html>

Lexemes can also be partially (un)inflectable, Type 2 in Table 1 (Baerman et al. 2015: 26–28; see Corbett, this volume, for a carefully documented typology of partial uninflectability): verbs of the English *hit* class have only -s and -ing inflections; Macedonian adjectives such as *kasmetlija* 'lucky, SG', *kasmetlii* 'lucky, PL' fail to show the expected three-way gender contrast in the singular (Friedman 1993). A particularly interesting case is that of a small group of neuter gender Latin loanwords into Polish with the endings -um or -o, *muzeum* 'museum', *studio* 'studio' (Swan 2002: 118; see Corbett, this volume, for detailed discussion). In addition to having the 'wrong' genitive plural form (*muzeów*, *studiów* for the expected zero ending form) these words are uninflectable in all the five otherwise distinct case forms of the singular 'slab' of their paradigm. Thus, while a native noun such as *pismo* 'document' has singular case marked forms *pismo*, *pisma*, *pismu*, *pismem*, *pismie* (for nominative/accusative/vocative, genitive, dative, instrumental, locative), *MUZEUM*, *STUDIO* have just *muzeum*, *studio*.¹

It is rarer for verbs to show partial uninflectability, but a form of this is seen in the Russian copular/verb of existence *BYT'*. In the past tense this verb is regular, formed on the root *by-* with the -l suffix of the 'l-participle' form, glossed *l* in examples (3).

- (3) a. U menja by-l-a kniga.
 at me[GEN] COP.PST-L-F.SG book(F).SG
 'I had a book.' [literally 'with me there.was a.book']
- b. Kniga by-l-a interesnaja.
 book(F).SG COP.PST-L-F.SG interesting.F.SG
 'The book was interesting.'

¹Loanwords borrowed as masculine gender nouns inflect normally, though they abstract out the -um component as the NOM.SG ending: *album*, *albu*, *albem*, *albie* 'album (nominative/accusative/vocative, genitive/dative, instrumental, locative singular)'.

In the present tense the copular/verb of existence is usually omitted altogether, (4a, b), but under certain circumstances it is possible to add the invariable present tense form *jest'*. In (4c) the overt copula adds rather subtle nuances of meaning so that (4c) tends to imply an individual-level or temporary predication, while the form without copula, (4a), tends to be interpreted as a stage-level or permanent state.

- (4) a. U menja kniga.
at me[GEN] book(F).SG
'I have a book.'
- b. Kniga interesnaja.
book(F).SG interesting.F.SG
'The book is interesting.'
- c. U menja jest' kniga.
at me[GEN] COP.PRS book(F).SG
'I have a book.'

Unlike all other Russian (non-impersonal) verbs, which agree with the subject in person/number, the present tense copular form is invariant for all persons/numbers, whether it is used purely as a copular, or whether it has an existential meaning. Example (5) is the title of a Russian pop song (https://www.youtube.com/watch?v=_oM4CQVXats), while example (6) is the title of a novel by V. Tokoreva.

- (5) U tebjja jest' ja.
at you[GEN] COP.PRS I[NOM.SG]
'You have me.'
- (6) Ja jest'. Ty jest'. On jest'.
I[NOM.SG] COP.PRS you[NOM.SG] COP.PRS he[NOM.SG] COP.PRS
'I am. You are. He is.'

The Russian copular, then, is an instance of a verb which inflects normally except in the present tense, where it is uninflectable for the expected person/number agreements.²

²In languages like Russian, Arabic and others in which the present tense copular is generally omitted altogether one might say that we have a maximally non-canonical instance of partial uninflectability, in which uninflectability manifests itself as total absence.

Russian *kenguru* is uninflectable in all its occurrences, a case of *lexical uninflectability*. However, English *kangaroo* is inflectable except in compounds: *kangaroo(*s) tails*, (arguably) an instance of *contextual uninflectability*, in which an otherwise inflecting lexeme has to be realized as a non-inflecting word form, or even as an uninflected bound stem, as in many cases of noun incorporation. Where the uninflectability seems to be an integral part of a grammatical construction, as in the case of English compounding, we can talk of *constructional uninflectability*, a subset of the contextual uninflectability type. When a lexeme becomes completely uninflectable because of the construction or context it is in we have an instance of Type 3, total constructional/contextual, uninflectability.

One type of constructional uninflectability closely related to English compounding is the Light Verb Construction (LVC) in certain languages (Spencer 2020). In most languages LVCs don't show (contextual) uninflectability, but straightforward uninflectedness. However, in some languages we find the same kind of contrast between inflectability and uninflectability in LVCs that we see in compounds in many languages. For instance, the well-known Japanese LVC formed with the empty verb *SURU* 'do' often alternates between a form with completely inert noun complement, as in *benkyoo suru* 'study' with bare noun *BENKYOO* 'studying', and a variant based on a grammatically marked form of the noun, *benkyoo o suru*, in which the noun complement of the light verb bears the object marker *o* (Miyamoto & Kishimoto 2016).

A particularly interesting case of contextual uninflectability is the predicative adjective in German, illustrated in (7).

- (7) a. Ich bin ein **kleines** Känguru.
 I am a little.N.SG.NOM/ACC kangaroo
 'I am a little kangaroo.'
- b. Das Känguru ist **klein**(*-e/*-es/...).
 the kangaroo is little(UNINFL)
 'The kangaroo is little.'

This type of uninflectability is found not just with predicative adjectives after the copula *sein*. It is found with secondary predication, postnominal attributive adjectives, and generally in any position other than canonical attributive modification, including adjective-noun compounds such as *Kleingeld* 'small change' (literally 'small_money', cf **Kleingeld*, **Kleinesgeld*), and we could also extend this to adverbial use (Trost 2006, Dürscheid 2002). For these reasons it is difficult to associate this contextual uninflectability with any specific construction type.

The German predicative adjective construction leaves an interesting, largely unexplored, question for paradigm-/lexeme-based models of inflection: precisely what grammatical features are borne by the predicative form of an otherwise inflectable German adjective? For instance, what number/gender/case feature values does the predicate adjective *klein* in (7b) actually bear, if any? And if it fails to bear any of the standard adjectival feature values, how is its form specified in the grammar, given that the bare root form *klein* itself never occurs in the position of an attributive modifier, but always bears some kind of inflectional suffix?

More generally, we can ask this question of any instance of contextual uninflectability, even those cases where it appears as though the uninflected lexeme has the form of an inflected word form in a cell of the lexeme's paradigm. Cetnarowska (2021) reports an interesting case in point. In Polish composite lexical units (a type of compound construction), an otherwise inflecting modifier word may appear as an uninflected element. She discusses cases such as *statek widmo* 'ghost ship', (8), formed by modifying the noun *statek* 'ship' with the (normally inflectable) noun *widmo* 'ghost'.

- (8) a. o statk-ach widm-o
 about ship-LOC.PL ghost-NOM.SG
 b. o statk-ach widm-ach
 about ship-LOC.PL ghost-LOC.PL
 'about ghost ships'

The examples (8a) and (8b) are identical in meaning and are equally acceptable. In example (8b) both components of the compound inflect for locative plural, but in (8a) the modifier noun *widmo* 'ghost' appears in the citation, nominative singular form. The phenomenon illustrated in (8a) is different from either the lexical uninflectability shown by Russian *kenguru*-words, or the contextual uninflectability of German predicative adjectives. The form *widmo* is, in fact, an inflected form, the nominative singular, part of the inflectional paradigm, and not an uninflected stem or root form.

Cetnarowska (2023) proposes a Construction Morphology (Booij 2010) solution to this paradoxical situation: take the N-*widmo* construction to be a kind of compound in which *widmo* is (optionally) specified to be in the nominative singular form only (taken to be the default form by Cetnarowska). In effect, this redefines one of the morphosyntactic/morphosemantic functions of the property set *nominative singular*. Another solution is to assume that when uninflecting, *widmo* is a distinct (though related) lexeme whose root form is specified as identical to the nominative singular form of the standard lexeme. This would have the

advantage of allowing us to code the metaphorical meaning change as a separate lexical property. However, there is little difference between the two analyses.

In some cases it is not immediately apparent whether we are dealing with lexical uninflectability or contextual uninflectability. Spencer (2020) discusses the case of indeclinable foreign names in Russian. Many borrowed or cited foreign words, and especially names, have a phonological shape which is difficult to accommodate to the Russian morphological system, like *kenguru* (no native nouns have a stem ending in a vowel) or *Si* (*Czin' pin*) 'Xí (Jìnpíng)'. Other foreign names, however, can easily be inflected as though they were Russian: *Bil* 'Bill', genitive singular *Bil-a* (cf *Kiril*, *Kiril-a*), or *Klinton* 'Clinton', genitive singular *Klinton-a* (cf *Anton* 'Anton', genitive singular *Anton-a*). However, native female referent names do not have phonological forms such as *Klinton*, and native given names do not end in *-i* for either sex, so that *Xilari Klinton* 'Hillary Clinton' is indeclinable, like *Si* 'Xí', (9).

(9) *reč'* 'the speech of ...'

- a. *Bill-a* *Klinton-a*
 Bill-GEN.SG Clinton-GEN.SG
 '...Bill Clinton'
- b. *Xilari* *Klinton*
 Hillary(UNINFL) Clinton(UNINFL)
 '...Hillary Clinton'
- c. *Xilari* *Benn-a*
 Hilary(UNINFL) Benn-GEN.SG
 '...Hilary Benn'

(Hilary Benn is a (male) British politician.) Not all foreign female referent names are indeclinable. Where the form of a name resembles that of a declinable Russian name it will be inflected, as in the case of the Spanish name in (10).

- (10) *knigi Teres-y* *Solan-y*
 books Teresa-GEN.SG Solana-GEN.SG
 'the books of Teresa Solana'

A different instance is that of the Russian form of the name of the Roman orator, Marcus Tullius Cicero, *Mark Tullij Ciceron*. All three components of this name are inflecting. In each case, Russian takes the bound stem of the Latin name and uses this as the unbound stem/nominative singular form, cf the original Latin genitive form *Marc-i Tulli-i Ciceron-is*.

Generally, there is no reason in principle why a female name like *Klinton* couldn't be inflected just like the male name but taking feminine gender agreements, in much the same way that thousands of male-referent names and kin terms (especially diminutives, such as *Andrjuša* from *Andrej*) inflect like feminine gender nouns but take masculine agreements: *Ja znaju èt-ogo mužčin-u* 'I know this-M.ACC/GEN.SG man-ACC.SG. Alternatively, Russian could have supplied female-referent names with a special feminine gender ending, like the Czech *-ová*. What this means is that we have to regard *Klinton*(M) and *Klinton*(F) as distinct, though related(!), lexemes, and we should analyse the female-referent names as an instance of total lexical uninflectability.

The partial lexical uninflectability seen with Polish *muzeum*-nouns is found more systematically in cases of "sporadic agreement" (Fedden 2019). In a number of Nakh-Dagestanian languages many verbs agree with (principally absolutive case) arguments. However, this agreement is often only shown by a minority of verbs. Other lexemes fail to show corresponding agreement, but otherwise show regular inflection for other verbal (TMA) categories. In this respect the verb system of these languages is similar to the prepositional lexicon of Celtic languages such as Welsh. Sporadic argument agreement is to be distinguished from differential object marking/agreement, where the presence of absence of agreement is semantically or pragmatically significant, for instance, indicating focus, definiteness, specificity and so on (see Corbett, this volume, for a detailed typology.)

The extant literature which is devoted explicitly to uninflectability has only clearly identified three distinct types: (i) lexically conditioned total uninflectability (Russian *kenguru*-nouns); (ii) lexically conditioned partial uninflectability (Polish *muzeum*-nouns); (iii) sporadic agreement. This raises the question of whether there might be clear cases of a fourth type, *partial contextual/constructional uninflectability*, in which all lexemes of a given class are expected to inflect in a given construction but in which some lexemes fail to inflect for a subset of features in those constructions. In other words can we find exemplars of the missing type in the simple typology shown in Table 2?

Table 2: Incomplete typology of uninflectability

	lexical	contextual
total	Type 1: <i>kenguru</i> -noun	Type 3: German predicative adjective
partial	Type 2: sporadic agreement	Type 4: ???

Polish *widmo*-compounds: The first candidate to consider is the Polish construction illustrated in (8a) above. The uninflecting element is, in fact, the nominative singular form, as we have seen, but that form is fixed by the construction itself. Though the *widmo*-compounds broadly fit the characterization of type (iv) uninflectability they are not perfect exemplars of that type, because only a relatively small number of lexemes seem to enter into the construction. It is therefore worth examining whether there are more general, lexically less restricted, cases of contextual partial uninflectability.

Latvian genitive compound: One candidate which is similar to the *widmo*-compounds but not lexically restricted is illustrated by a class of noun-noun (N N) compounds in Latvian. In this language, a very common way for a noun to serve as attributive modifier to a head noun is through a noun-noun headed compound construction type in which the modifier noun obligatorily appears in the genitive case, either singular (11a) or plural (11b) (Mathiassen 1997: 56).

- (11) a. kafij-as tasīte
 coffee-GEN.SG cup
 ‘coffee cup’
 b. latvieš-u valoda
 Latvian-GEN.PL language
 ‘Latvian (language)’ [literally ‘language of the Latvians’]

Although Latvian has NN compounds in which the modifier is in the root form, as well as relational adjectives, the N_{genitive} + N construction has a high token frequency. It is much more common than, say, the corresponding construction in the closely related Lithuanian. Thus, while ‘stone house’ is normally rendered with the N_{genitive} + N construction in Latvian, in Lithuanian a relational adjective would be found, agreeing like any other adjective in number, gender, case (Kalnača & Lokmane 2016: 174).

- (12) a. Latvian
 akmen-s nams
 stone-GEN.SG house(M).NOM.SG
 ‘stone house’
 b. Lithuanian
 akmen-in-is namas
 stone-REL.ADJ-M.GEN.SG house(M).NOM.SG
 ‘stone house’ [literally ‘petrous house’]

In Latvian, many of the N_{genitive} + N expressions, the genitive attributive noun is itself modified, by nouns, adjectives, numerals, prepositions and other elements (Kalnača & Lokmane 2016: 171–172).

- (13) a. liel-zied-u cerīni
large-flower-GEN.PL lilac
'large-flowered lilac'
- b. deviņ-stāv-u māj-a
nine-storey-GEN.PL house
'nine-storeyed house'
- (14) a. vien-zilb-es vārds
one-syllable-GEN.SG word
'monosyllabic word'
- b. bez-maks-as pakalpojums
without-charge-GEN.SG service
'free service'

In a number of such cases the compound N_{genitive} modifier is modified by a preposition which would normally govern a case other than the genitive, comparable to ‘paratactic’ adjectives in other European languages, such as Russian *za-rubežnyj* ‘overseas’, from *za* ‘beyond’, *rubež* ‘border’, *-nyj* ‘adjective-forming suffix’. The difference is that in Latvian the noun remains a noun and does not have to be transposed into a (relational) adjective. The structure is thus akin to that of the English *over-seas*. In the Latvian construction the noun remains in the genitive case, irrespective of what case is governed by the compounded preposition (Kalnača & Lokmane 2016: 181). This shows that the genitive has been frozen as an exponent of the modifier function of the noun. For instance, when governing a noun phrase in the plural, all Latvian prepositions take dative case (whatever case they govern in the singular), but in the corresponding compound genitives the noun remains in the genitive plural, (15–16). The $N_{\text{genitive}} + N$ constructions, including the compound genitives can also function as nominal predicates, (17).

- (15) a. N + P + DAT
 tablet-es pret sâp-ēm ⇒
 pill-NOM.PL against pain-DAT.PL
 ‘pills against pain’
 b. compound genitive
 pret-sâpj-u tablet-es
 against-pain-GEN.PL pill-NOM.PL
 ‘painkillers’

- (16) a. N + P + DAT
autobus-s starp pilsēt-ām ⇒
bus-NOM.SG between city-DAT.PL
‘bus between cities’
- b. compound genitive
starp-pilsēt-u autobus-s
between-city-GEN.PL bus-NOM.SG
‘intercity bus’
- (17) Šis pakalpojums būs bez-maksas.
this.NOM.SG.M service(M)NOM.SG COP.FUT.3 without-charge.GEN.SG
‘This service will be free.’ (Kalnača & Lokmane 2016: 178)

The attributive genitive marked noun can also be used in elliptical contexts which represent a non-canonical form of attributive modification. Thus, a set of teaching materials by Nau (2008) entitled *Gramatikas modulis* (‘grammar-GEN.SG module’), has a Latvian-Polish glossary with the title (18).

- (18) student-u vārdnīca (latvieš-u – pol-_u-u)
student-GEN.PL vocabulary (Latvian-GEN.PL Pole-GEN.PL)
‘students Latvian-Polish vocabulary’

Here, the genitive plural forms *latvieš-u* ‘Latvian’ and *pol-_u-u* ‘Polish’ function as attributive modifiers to an elided head noun *vārdnīca* ‘dictionary’.

The conclusions from the Latvian genitive attributive construction are that genitive case functions as a kind of ‘frozen’ inflection, much like the nominative singular form of Polish *widmo*-nouns. However, the attributive nouns retain the number contrast, so they are not completely uninflectable. In this respect they are akin to Polish *muzeum*-nouns, except that they are constructionally uninflectable, not lexically uninflectable. The Latvian genitive compounds therefore not only present a plausible example of the missing type (iv) of uninflectability, but also highlight the point made by Corbett (this volume), that uninflectability is not a binary or absolute category, but admits of degrees of variation.

Hungarian inflecting infinitive: Another possible instance of type (iv) uninflectability is the Hungarian inflecting infinitive. The infinitival complement to certain auxiliary(-like) verbs, *kell* ‘need’, *lehet*, *szabad* ‘is possible/allowed’, inflects like a possessed noun, shown in boldface in (19), agreeing with the (understood) subject in person/number ((19a), Kenesei et al. 1998: 314, (19b), Kiss 2002: 210, (19c), Kenesei et al. 1998: 318).

- (19) a. Nekem most men-n-**em** kell.
me.DAT now go-INF-1SG.PX need
'I have to go now.' [present tense]
- b. Korábban kell-ett volna neked haza men-n-**ed**.
earlier need-PST CONDITIONAL you.DAT home go-INF-2SG.PX
'You needed to have gone home earlier.' [past conditional]
- c. Nem fontos magyarul beszél-ni-**e**.
not important Hungarian speak-INF-3SG.PX
'It isn't important that s/he speak Hungarian.' [complement to *fontos*]

With other (auxiliary-like) verbs, there is no inflection, (20), (Kenesei et al. 1998: 314).

- (20) Nem tud-ok/bír-ok fel-áll-ni.
not able-1SG.PRS up-stand-INF
'I am not able to stand up.'

No inflection is found with unspecified subjects, presumably because as far as the verb is concerned, there is no subject, (21), (Kenesei et al. 1998: 314).

- (21) Mit kell a kerttel csinál-ni?
what need the garden.to do-INF
'What needs to be done to the garden?'

As in the case of German predicative adjective uninflectability, it is not clear what counts as a 'construction' here, so it is safer to label the phenomenon as 'contextual uninflectability'.

The Hungarian inflecting infinitive is the constructional counterpart to the sporadic agreement phenomenon found in Nakh-Dagestanian languages and elsewhere. The difference is that it is not lexically conditioned, it is determined by the wider syntactic context and applies to all appropriate lexemes. Together with the Latvian genitive compounds, the Hungarian (partially) inflecting infinitive serves to fill in the missing cell in the partial typology of uninflectability given in Table 2, allowing us to propose the typology in Table 3.

3 What counts as inflection?

We have so far been tacitly assuming that we know what it means for a lexeme to inflect. However, any serious study of the nature and typology of uninflectability must first establish what exactly is meant by *inflection*. There are

Table 3: Four-way typology of uninflectability

	lexical	contextual
total	<i>kenguru</i> -noun	German predicative adjective
partial	<i>muzeum</i> -noun	Latvian genitive compounds (Polish <i>widmo</i> -compounds)
	sporadic agreement	Hungarian inflected infinitive

several types of morphological or morphosyntactic process that are closely related to inflection, and which in some ways could be said to count as inflection, so it is worthwhile asking whether an uninflectable lexeme can undergo those processes. These include clitic-like affixation (or conversely, affix-like clisis), periphrasis, and various types of evaluative morphology, for instance, diminutive formation. (Admittedly superficial) examination of extant descriptions of uninflectable lexemes suggests that none of these processes count as inflection for the purposes of defining uninflectability. In other words, uninflectability only seems to relate to what we might call ‘core’ inflection: uncontroversial affixation or stem modification serving to realize a central repertoire of functional categories (hence, excluding evaluative morphology).

According to some authors, pronominal and other elements that are traditionally referred to as clitics in a variety of languages should really be thought of as a particular type of affix, which effectively means that the properties those ‘clitics’ realize are inflectional features (e.g. Monachesi 1999 for Italian, Joseph 1988 for Greek, Lyons 1991 for Macedonian, Miller & Sag 1997, Bonami & Boyé 2007 for French amongst others; see also Spencer & Luís 2012). The uninflectable French verb FOLLOW, however, seems to cooccur freely with these bound pronominals in (2a), suggesting that such clitic-affixes do not count as inflection for the purposes of uninflectability.

Similarly, it has been claimed that the postpositive definite articles in Bulgarian and Macedonian are better thought of as phrasal affixes or edge inflections, rather than straightforward clitics (Sadock 1991, Halpern 1995; see also Spencer & Luís 2012: 129–30). But these, too, can attach to uninflectable adjectives in those languages, e.g. Macedonian *taze-to* ‘fresh-DEF’ ‘the fresh (one)’ (multiple attestations on the internet).

In (2b) we see an instance of an uninflectable French verb licensing the periphrastic past tense (‘passé composé’) construction, suggesting that periphrasis, too, fails to count as core inflection (Spencer & Popova 2015: 208).

The status of evaluative morphology with respect to traditional notions of inflection and derivation is notoriously unclear (for discussion see Spencer 2013: 113–122; Grandi 2015). I shall adopt the view that certain types of evaluative morphology have to be distinguished from derivation or word-formation proper, that is, from lexeme formation. We can encapsulate this stance as the *Novel Lexeme Criterion*: If a morphological process applies to some form of a lexeme and can only sensibly be said to deliver a form belonging to the same lexeme then that process is not one of derivation/word-formation. A case in point would be diminutive/augmentative forms of proper names. In Russian, a name such as *Andrej* can give rise to a number of expressive diminutive forms such as *Andrjuša*, *Andrjušen'ka*. However, these diminutive forms will all denote the same individual as that denoted by the more formal variant *Andrej*. In this sense, the diminutives are just forms of the basic name. (This does not necessarily mean that it makes sense to call diminutive formation an inflectional process, of course.)

The first point to establish is that evaluative morphology can often be applied to uninflecting words, that is, words which belong to classes which are not expected to take any kind of inflection in any case. The most obvious such instances are seen in languages with no or virtually no inflectional morphology in the first place. Chinese, for instance, lacks (uncontroversial) inflections, but has non-affixal evaluative morphological processes: reduplication, ablaut, tone shift (Arcodia 2015). In inflecting languages, we find that evaluative morphology can be applied to uninflecting word classes such as adverbs or quantifiers. This is true of the Basque diminutive suffix *-txo* (Artiagoitia 2015: 197–198), and the Jingulu evaluative suffix *-kaji* (Pensalfini 2015: 416).

The propensity for evaluative morphology to apply to uninflecting words, even if with strong lexical limitations, seems rather widespread. Even in Russian, for instance, which has adjective-based evaluative morphology in addition to very productive noun-based evaluation, but no productive evaluative morphology for other parts of speech, we find isolated instances such as the diminutive form of the word *spasibo* ‘thank you’ $\Rightarrow$ *spasib-k-i* (diminutive suffix *-k-* with an apparent plural ending *-i*). Given this background we can ask what potential an uninflecting name in Russian might have for diminutive formation. In my extended family there is a dog whose English name is Harry, but who also responds to the Russian form of the name, *Xari*. The Russian form is uninflectable. However, it readily takes the diminutive form *Xari-k* (which is how he is usually addressed by Russian-speaking family members). If the set of situations just described are typical then it indicates that evaluative morphology, too, does not count as *inflection* for the purposes of uninflectability.

Uninflectable lexemes can sometimes undergo derivation-like inflectional processes. For instance, comparative/superlative formation in Indo-European adjectives and adverbs does not seem to create a new lexeme, yet it adds an important meaning component to the original base word and changes its syntactic potential considerably. In this respect it is far from canonical as inflection. It is therefore not too surprising that the German contextually uninflectable predicate adjectives have comparative forms, (22).

(22) German

- a. Das Känguru ist klein.
the kangaroo is little(UNINFL)
'The kangaroo is little.'
- b. Aber dieses ist noch klein-er.
but this-one is even little-COMPARATIVE
'But this one is even smaller.'

It should equally come as no surprise to learn that uninflectable lexemes can undergo uncontroversially derivational morphology. More significantly, such lexemes can undergo morphology which sits on the boundary between canonical derivation and canonical inflection. In Table 4 we see examples of standard derivation (*konoščnik*, *kengurjonok*), and the formation of what Spencer (2013) refers to as a transpositional lexeme, in this case a relational adjective that adds no semantic predicate to the base noun but which nevertheless behaves like an autonomous lexeme, *kofejnyj* (for a justification of this claim and an extensive crosslinguistic typology of relational adjectives see Nikolaeva & Spencer 2019).

Table 4: Derivation applied to uninflectable Russian nouns

<i>kino</i>	'cinema'	⇒	<i>konoščnik</i>	'cinema worker'
<i>kenguru</i>	'kangaroo'	⇒	<i>kengurjonok</i>	'joey (baby kangaroo)'
<i>kofe</i>	'coffee'	⇒	<i>kofejnyj</i>	'pertaining to coffee' (relational adjective)

Finally, uninflectable lexemes may undergo compounding, including the more morphological type of compounding which involves an intermorph, as in the Russian examples shown in (23), where *klin* 'wedge' is an ordinary inflecting noun while *kino* 'cinema' is uninflectable.

- [illegible]

4 A paradigm-based solution

We turn now to the problem of how we can best represent the various types of uninflectability in a formal model of morphology. This is a non-trivial problem for at least two reasons. First, as we have seen, and as Corbett (this volume, Section 4) demonstrates at length, uninflectability turns out to be a multi-dimensional phenomenon which is best thought of in terms of degrees of uninflectability rather than an absolute property. Second, there are a number of conceivable ways of stating the facts of uninflectability in various frameworks. However, we should aim to go beyond simple description and strive towards the more interesting goal of stating those facts in a way that reveals some kind of insight into the underlying nature of (the various types of) uninflectability, and which might thus provide the beginnings of an explanation for the observed patterns.

Total uninflectability is a relatively neglected problem, especially in the theoretical literature. I have already mentioned the important monograph-length descriptive study by Doleschal (2001), but uninflectability is not mentioned in Stump (2016), nor can I find reference to it in recent handbooks such as Baerman (2015), including psycholinguistic studies (Stoll 2015, Walenski 2015). The discussion in Brown & Hippisley (2012: 65, 156–159, 162), in which uninflectedness is treated as a kind of syncretism with the root form, is an honourable exception (see also Baerman et al. 2017: 26–28.).

In this section I will develop a formal account based on a realizational model of inflection combined with the approach to lexical representation advocated in Spencer (2013). It is important to be able to propose a formal account that encompasses all four types in the typology of Table 2 in essentially the same way. Otherwise, we will have no guarantee that we are not dealing with apparently similar but in fact disparate phenomena. What this means is that there has to be some core formal property or collection of properties that is shared uniquely by the four types of uninflectability but not by other phenomena.

Traditional morpheme-based models (including classical LFG) would presumably be obliged to postulate up to twelve zero suffixes for *kenguru*-nouns and

for the Russian (Hillary) *Klinton*. Distributed Morphology (Embick 2015) might be able to deploy the device of Impoverishment to neutralize formal distinctions between otherwise inflected forms by retreating to some maximally unmarked form identical to the lexical root. But it is hard to see how this would work for languages such as Greek, in which all inflected words are affixed forms of a bound stem (cf the Latin noun declension illustrated below in Table 9). For such cases it would presumably be necessary to posit a special zero suffix for some arbitrarily chosen cell/feature set and then Impoverish the rest of the paradigm to that cell. Uninflecting lexemes thus present a particularly severe challenge to such models. Inferential-realizational models such as PFM1 Stump (2001) would have to posit massively neutralizing rules of referral (always to the lexical base form), not much of an improvement.

Spencer (2020) argues for an analysis which makes crucial appeal to the distinction between two types of paradigm proposed in Stump (2016), in a model that has come to be known as PFM2. PFM2 distinguishes crucially between content and form paradigms (Π_C , Π_F). The leading idea is very similar to that which motivates the distinction between s(yntactic)-features and m(orphological)-features in Sadler & Spencer's (2001) analysis of the Latin perfect passive periphrasis. The s-features are those features that govern syntactic constructions and which may receive some special non-lexical semantic interpretation. For example, in English definiteness is a property of noun phrases which gives rise to (very subtle and poorly understood) semantic and pragmatic contrasts. It is also an obligatory property, much like a purely inflectional property such as verb tense. However, it is never expressed by a purely morphological form, rather it is expressed by specific function words (in/definite articles, demonstratives) and other constructional properties (e.g. possessive phrases). This means that it is a purely syntactic feature. In Scandinavian languages, for instance, Danish (Lundskær-Nielsen & Holmes 2015) definiteness is expressed both by function words and by special inflected forms of the noun, while in Latvian (Kalnača & Lokmane 2021) definiteness is expressed by special morphological forms of attributive adjectives. For those languages, then, definiteness is a feature which regulates not only syntactic structure, but also morphological form.

In PFM2 Π_C defines all the syntactically accessible inflectional contrasts a lexeme is obliged to make. It serves as the interface between morphological forms and syntactic-semantic structures, in the sense that the cells of the Π_C are defined by features which represent all the morphosyntactic contrasts visible to the syntax, as well as all the functional semantic contrasts that are interpreted by the semantics. This is Stump's (2016: 105) *interface hypothesis*: "Paradigms are the interfaces of inflectional morphology with syntax and semantics". Given this

characterization, the features, σ , defining the Π_C are a subset of the s-features, the total set of features required to specify syntactic structures and the interpretation of functional features (as opposed to lexical meanings).

By contrast, the form paradigm, Π_F , serves as part of the definition of the set of morphophonological forms expressing the contrasts defined at the level of the Π_C (Stump 2016: 106–115). The Π_F for a lexeme $\mathcal{L}$ is a set of pairings of complete property sets³ with a set of appropriately selected stems for $\mathcal{L}$. The cells in Π_F are therefore ordered pairs of the form $\langle Z, \tau \rangle$, where Z is a member of the lexeme $\mathcal{L}$'s stem set and τ is a set of (morphomic) morphosyntactic properties possibly distinct from the set σ . It is Π_F that serves as the point of departure for the application of the rules of morphology, the Paradigm Function, which maps the cells of Π_F to a realized paradigm (Π_R), consisting of pairs of the form $\langle \omega, \tau \rangle$, where ω is the inflected word form (or a set of word forms in the case of overabundance, Thornton 2012). In this respect the Π_F is a theory internal construct whose role is to serve as the input to the battery of realization rules that make up the PFM2 morphological engine.

The paradigms Π_C and Π_F are related through a principle of Paradigm Linkage, mediated by *Corr*, a Correspondence function which specifies the mapping $\Pi_C \mapsto \Pi_F$, in part defined by the function *pm* ('property mapping') defined over the feature sets, Σ , T , of Π_C , Π_F . The Π_C of a lexeme is thus a matrix of cells consisting of ordered pairs of the form $\langle \mathcal{L}, \sigma \rangle$, where σ is some subset of Σ . The component $\mathcal{L}$ is a lexical index which identifies an equivalence class of lexemes by their inflectional paradigm, ignoring irrelevant homophony/homonymy where appropriate. In this respect, a word such as the English verb *DRAW* belongs to a single lexemic equivalence class, even though it has a wide range of meanings, some of them totally unrelated to each other (*draw a tooth*, *draw a conclusion*, *draw a picture*, ...). In other words, the lexemic equivalence class with respect to morphology is defined by the shared inflectional paradigm. This is not the only way that lexemes need to be individuated, of course. As Fradin & Kerleroux (2003) point out, derivational morphology often targets very specific meanings of homonymous or polysemous lexemes, even if those lexemes share all their inflectional features (see also Bonami & Cysmann 2017 for further discussion). For the cases I shall be discussing, however, only very coarse grained distinctions between lexemes are required. The component σ is a complete and coherent morphosyntactic property set, which the syntax-semantics needs to have access to, so as to express purely syntactic dependencies such as agreement and govern-

³Stump refers to 'properties' rather than features. For our purposes we can treat these terms as synonymous, but I use 'property' when discussing Stump's own proposals.

ment, and for realizing or specifying the values of obligatory meaningful categories such as nominal number, verb tense and so on. The set σ crucially does not include purely morphological (morphomic) properties such as inflectional class indices.

In a canonical inflectional system the Π_C and Π_F are isomorphic: each lexeme $\mathcal{L}$ is associated with a unique root, Z , its sole inflectional stem, and $\tau = \sigma$, and $Corr$ and pm are the identity function. This is canonical paradigm linkage as defined in Stump (2016: 113, example (2)). Note that the properties which define Π_C and Π_F are taken from the same vocabulary of properties and in the canonical case the two sets are identical, not *merely* isomorphic, but actually identical (Stump 2016: 113; see below). However, Stump (2016) catalogues a variety of non-canonical situations, in which $\tau \neq \sigma$. The deviations from the canonical isomorphic mapping between Π_C and Π_F include syncretism, deponency, heteroclisis, overabundance, defectiveness. In a number of cases the deviation from canonicity is purely morphomic. Heteroclisis, for instance, is found when a lexeme inflects according to more than one inflectional class in different parts of its paradigm (say, singular vs plural forms). In the case of deponency, e.g. Latin verbs which are ‘passive in form but active in meaning’, we have an unexpected mapping from the content property ‘active voice’ to a set of forms which would normally realize passive voice. In the case of syncretism we have a pair of contrasting Π_C properties which map to identical Π_F properties, neutralizing the Π_C property contrast. For instance, Stump (2016: 174–75) handles a set of syncretisms in the Bhojpuri verb by ‘deleting’ certain property values from the Π_F paradigm in the pm and hence halving the number of cells.

In a few of the case studies presented in Stump (2016), notably the Nepali ‘long present negative’ conjugation (following the analysis proposed in Bonami & Boyé 2008) the set τ includes properties which Stump (2016: 139–145) describes as ‘morphomic’. These properties, which Stump, following Bonami and Boyé labels ‘row’ and ‘column’, have no correspondent at all in the Π_C . However, in the great majority of cases the properties in the set τ are formally exactly the same objects as the properties in σ . This majority situation reflects the property-set preservation principle as described by Stump (2016: 113), (24).

- (24) Property-set preservation: Content cells and their form correspondents have the same property set; that is, for any property set σ associated with a content cell, $pm(\sigma) = \sigma$.

Property-set preservation conceals a mild conceptual inconsistency in PFM. That model is based on the principle of the autonomy of morphology or *morphology-by-itself* (Aronoff 1994), as discussed extensively in Stump (2016: Chapter 8):

“... morphomic properties can all be seen as an effect of properties that are restricted to the level of form paradigms” (Stump 2016: 120). But the reverse should hold too, given the architectural assumptions of PFM: all the properties of Π_F (and Π_R) should themselves be morphomic, for the simple reason that neither syntax nor semantics have access to those levels of representation (cf Boyé & Schalchli 2019; Spencer 2021). Exploring these issues goes beyond our remit here.

I will illustrate the virtue of the Π_C/Π_F distinction with a brief survey of German noun declension. This case illustrates the extent to which inflection can be morphomic but more importantly for our present purposes it also illustrates the kind of case which might be mistaken for a case of (partial) uninflectability when in fact it is something else.

In standard descriptions, German nouns inflect for four cases and two numbers (Durrell 2002), so that the German noun phrase is defined by an 8-celled paradigm, illustrated in Table 5 by the declension of the definite article in the masculine singular form. In Table 6 we see the declension of two typical nouns, *VATER* ‘father’, and *HAND* ‘hand’, and in Table 7 we see the declension of the ‘weak’ noun *STUDENT* ‘student’. Nouns of the *Student*-class, traditionally called weak nouns, differ from so-called ‘strong’ nouns in that all their forms end in *-en* except for their nominative singular form. We might be tempted to assume that weak nouns are therefore indeclinable for 7/8ths of their paradigm and only inflect (subtractively) in the nominative singular. However, this would be to completely misanalyse German noun inflection.

Table 5: German definite article masculine singular

	SINGULAR	PLURAL
NOMINATIVE	der	die
ACCUSATIVE	den	die
GENITIVE	des	der
DATIVE	dem	den

Let us call the features defining the Π_C ‘ σ -features’ (a subset of the s -features) and those defining the Π_F ‘ μ -features’ (for ‘morphological/morphomic’ features, broadly equivalent to m -features). In German there is a considerable mismatch between the σ -feature set required for the Π_C and the μ -feature set (Boyé & Schalchli 2019; Spencer 2009). What is unusual about the weak nouns is that they show that the morphomic feature inventory depends on lexical class. In the strong nouns the inflectional split is between singular and plural subparadigms

2 Uninflectability: Concepts and consequences

Table 6: Standard description of German ‘strong’ noun declension: VATER, HAND

	VATER ‘father’		HAND ‘hand’	
	SINGULAR	PLURAL	SINGULAR	PLURAL
NOMINATIVE	Vater	Väter	Hand	Hände
ACCUSATIVE	Vater	Väter	Hand	Hände
GENITIVE	Vaters	Väter	Hand	Hände
DATIVE	Vater	Vätern	Hand	Händen

Table 7: Standard description of German ‘weak’ noun declension: STUDENT

	STUDENT ‘student’	
	SINGULAR	PLURAL
NOMINATIVE	Student	Studenten
ACCUSATIVE	Studenten	Studenten
GENITIVE	Studenten	Studenten
DATIVE	Studenten	Studenten

(mainly), rather than nominative singular and the rest of the paradigm. Notice that the STUDENT class is not the only one whose paradigm only has two distinct forms. The same is true of MÄDCHEN ‘girl’, WOHNUNG ‘flat, apartment’, PARK ‘park’, Table 8.

Table 8: Standard description of German noun declension: MÄDCHEN, WOHNUNG, PARK

	MÄDCHEN ‘girl’		WOHNUNG ‘apartment’		PARK ‘park’	
	SG	PL	SG	PL	SG	PL
NOM	Mädchen	Mädchen	Wohnung	Wohnungen	Park	Parks
ACC	Mädchen	Mädchen	Wohnung	Wohnungen	Park	Parks
GEN	Mädchens	Mädchen	Wohnung	Wohnungen	Parks	Parks
DAT	Mädchen	Mädchen	Wohnung	Wohnungen	Park	Parks

In fact, no German noun distinguishes more than four of the eight theoretically possible forms (the noun *BEAMTE* ‘civil servant’ and nouns such as *ANGESTELLTE(R)* ‘employee’ inflect like adjectives, not nouns). The ‘weak’ nouns are the only nouns which ever distinguish nominative from accusative case and then only in the singular (a small set of masculine irregular nouns such as *NAME* ‘name’ also distinguish nominative singular from accusative singular, but they only have a total of three distinct forms each in their complete paradigm: *Name*, nominative singular, *Namens*, genitive singular, *Namen* elsewhere). It is therefore more in keeping with our morphomic assumptions to characterize the German noun Π_F by just giving arbitrary labels to the various form classes, as illustrated in (25), where I conflate in part the (abstract) Π_F with the concrete realized paradigm.

- (25) a. Form1=Vater; Form2 = Form1 $\otimes$ s; Form3=Umlaut(Form1);
Form4=Form3 $\otimes$ n
b. Form1=Hand; Form2 =Umlaut(Form1) $\otimes$ e; Form3=Form2 $\otimes$ n
c. Form1=Mädchen; Form2 = Form1 $\otimes$ s
d. Form1=Wohnung; Form2 = Form1 $\otimes$ en
e. Form1=Park; Form2 = Form1 $\otimes$ s
f. Form1=Student; Form2 = Form1 $\otimes$ en

Notice that these specifications give the superficial impression that *WOHNUNG* and *STUDENT* belong to exactly the same lexical class, but that impression is dispelled when we look at the *Corr* function. This specifies Form2 for the *Wohnung*-class as the Π_F correspondent of Π_C [NUMBER:plural], while for the *Student*-class, Form2 is the default correspondent and Form1 is specified as the Π_F correspondent of [NUMBER:singular, CASE:nominative]. Sample rules are provided in (26).

- (26) (i) Class:VATER
[NUMBER:plural, CASE:dative] $\Rightarrow$ Form4
[NUMBER:plural] $\Rightarrow$ Form3
[NUMBER:singular, CASE:genitive] $\Rightarrow$ Form2
Elsewhere $\Rightarrow$ Form1
(ii) Class:STUDENT
[NUMBER:singular, CASE:nominative] $\Rightarrow$ Form1
Elsewhere $\Rightarrow$ Form2

- (iii) Class:WOHNUNG
 [NUMBER:plural] $\Rightarrow$ Form2
 Elsewhere $\Rightarrow$ Form1⁴

What we learn from systems like the German noun paradigm is that there is no necessary simple correspondence between inflectional form and inflectional function. But more than that, the divergence between form and function can become so great that a lexeme hardly appears to inflect at all. The German weak nouns are just one step away from becoming uninflectable. This illustrates yet again the point made by Corbett (this volume, especially Section 4) that uninflectability is not an all-or-nothing affair. It is then the task of the formal grammarian to find adequate ways to represent this fact.

We are now in a position to propose a formal model of uninflectability within a paradigm-based morphology. The leading intuition will be that an uninflecting lexeme has a full Π_C but a depleted or even non-existent Π_F . Recall that the original motivation for the $\Pi_C \sim \Pi_F$ distinction is the idea that the Π_C represents the interface with syntax/semantics, while the Π_F represents the interface with (morpho)phonology. One way of thinking of this is to say that PFM2 is a model of *lexical insertion*: the Π_C defines a set of properties that a lexeme is obliged to realize in a given language, while the Π_F specifies what kinds of morphophonological forms will be available to realize those properties. In other words, the *Corr* function regulates the syntax/semantics~morphology interface by specifying which Π_F cell occupant corresponds to a given Π_C feature set. In this way, *Corr* regulates lexical insertion for inflecting lexemes (cf Stump 2016: 27–30, 115).

Central to the argument, then, is the claim that the problem of uninflectability is intimately related to the question of lexical insertion. We can reframe the problem in this way: how can a lexical model of inflection license the insertion of lexical forms which apparently fail to bear inflection? We must therefore address the question of how lexical insertion is conceived in a paradigm-/lexeme-based model of morphosyntax. We begin with the most innocuous looking of lexical elements, the uninflecting lexeme.

The uninflecting lexemes of a language, such as English prepositions, can easily be defined as members of a lexical class which is not associated with any Π_C . In the architecture of PFM2 the cells of the Π_F are defined by projection from Π_C by means of the *Corr* mapping. Thus, in the absence of a Π_C the Π_F will be undefined. How, then, can we determine the form such a lexeme takes when it is selected to be part of a sentence (lexical insertion)?

⁴Choice of Form1 as the default is arbitrary, of course, mandated only by the fact that singular tends to be the default number in languages like German.

One possibility is that we treat uninflecting lexemes as having a trivial inflectional paradigm, consisting of one Π_C cell, $\langle \mathcal{L}, \sigma \rangle$ mapped to a Π_F cell $\langle Z, \emptyset \rangle$, where $\emptyset$ is a null form property of some kind. The feature $\emptyset$ is not associated with any realization rule, so we still have no way of specifying the final form of the inserted lexeme. The obvious step would be to invoke Stump’s Identity Function Default (IFD). However, it is not clear how this could be done given the standard PFM architecture.

The motivation for the IFD is to provide an output for a maximally general, underspecified input. However, this application is paradigm-driven, in the following sense. The inflectional paradigm of any lexical class is defined in terms of some ensemble of features. Consider the simplest case of a paradigm with just two cells, say, a noun inflecting for $\text{NUMBER}:\{\text{singular}, \text{plural}\}$. The IFD guarantees that the cell, say, $\langle \text{DOG}, [\text{NUMBER}:\text{singular}] \rangle$ will be mapped to a Π_F correspondent, even in the absence of a realization rule defined over the feature specification $[\text{NUMBER}:\text{singular}]$, namely, the form given by the stem selection rule for English nouns, which itself by default delivers the lexical root form $/\text{dog}/$.

Implicit here is the assumption that inflectional defectivity is to be avoided, *ceteris paribus*. The grammar demands that an English count noun have an inflectional paradigm with two forms, one for singular and one for plural (even if those forms are morphophonologically identical, as in the case of *SHEEP*). The IFD guarantees that *DOG* has a singular form, by applying the totally underspecified realization rule (27) in the first (sole) inflectional rule block, where $X = /dog/$ (cf. Bonami & Stump 2016: 465).

$$(27) \quad X_U, \{ \} \rightarrow X$$

Rule (27) states that a form ‘X’ (whether basic or complex), realizes a completely underspecified morphosyntactic property set $\{U\}$, by being mapped to itself.

Table 9: Paradigm of Latin *DOMINUS* ‘lord’

	SINGULAR	PLURAL
NOMINATIVE	domin-u-s	domin-i:
VOCATIVE	domin-e	domin-o:-s
GENITIVE	domin-i:	domin-o:-rum
DATIVE	domin-o:	domin-i:-s
ABLATIVE	domin-o:	domin-i:-s

In some cases it will appear that the IFD fails to apply. Latin nouns inflect for five cases and two numbers. A 2nd declension noun such as DOMINUS ‘lord’ has a root form /domin/ which is bound and thus never appears in an unsuffixed form, as can be seen from Table 9. Even if we assume that its inflected forms are based in part on a derived stem form (say, *domin-u* or *domin-o*), there will be no cell in the noun’s paradigm that is not formed with an inflectional suffix, so that the whole paradigm has to be defined by the application of non-trivial, specified realization rules, including the nominative singular form. When applied to a lexeme such as DOMINUS the IFD would define a root form *domin* (and possibly also an uninflected bound derived stem *domin-u*, *domin-o* or whatever, depending on the details of the analysis). However, such a bound form would never be realized as a fully inflected word form, and would thus never be eligible for lexical insertion into a syntactic phrase, because all the cells in the paradigm of DOMINUS are filled by forms which are defined by inflectional rules which are more specific, and which thus pre-empt the IFD.⁵

Given the architecture of PFM it is clear that the IFD is irrelevant to the question of how to specify the form of an uninflecting lexeme for lexical insertion, for the simple reason that the IFD specifies forms in inflection paradigms, but an uninflecting lexeme, by definition, has no inflectional paradigm, no paradigm function, and a fortiori, is not subject to any realization rule, not even the maximally underspecified rule (27) which instantiates the IFD.⁶

The only way we could deploy the IFD for uninflecting lexemes, it seems, is to provide such lexemes with a paradigm. This would entail creating a special inflected form, realizing some property, say, [LEXICALINSERTION:yes], whose inflectional (one-celled) paradigm is not specified by any stipulated realization rule. This would mean that its form is invariably defined by the IFD. In other words, each uninflecting lexeme would have a Π_C whose cell realized the [LEXICALINSERTION:yes] feature, and whose form correspondent was $\iota(Z)$, where ι is the identity function and Z is the lexical root. In fact, given the assumptions of PFM2, such a treatment is the only one possible. The Π_C , recall, represents the interface between morphological form (the contents of the Π_F cells) and syntax-semantics. Therefore, either we have to assume that uninflecting lexemes do, after all, have a Π_C , and are hence “covertly” inflecting, or we have to conclude

⁵The inflectional rules of Latin can be expected to take nominative (and singular) as default values of the CASE and NUMBER features, in which case there would be an important sense in which the IFD *does* apply even to DOMINUS. It is just that its effects are hidden.

⁶This conclusion is equally valid for the morph-based interpretation of PFM proposed by Crysmann & Bonami (2016) and Crysmann (2020) under the rubric of Information-based Morphology (IbM).

that uninflecting lexemes fall outside the PFM architecture in its current form, and that they interface with the rest of the grammar in some other way.

The ploy of creating a ‘dummy’ inflectional paradigm for uninflecting lexemes is, in essence, is the solution proposed by Sag (2012: 119). He assumes a “trivial” *Zero Inflection Construction*, shown in (28), “which pumps noninflecting lexemes to word status. This construction takes a lexeme as daughter, licensing mothers that are of type *word*, but otherwise identical to their daughter.”

$$(28) \quad \text{Zero Inflection Construction } (\uparrow \text{infl-cxt}) \\ \text{zero-infl-cxt} \Rightarrow \left[\begin{array}{l} \text{MTR } X! \text{ word} \\ \text{DTRS } \langle X : \text{invariant-lxm} \rangle \end{array} \right]$$

The type feature *invariant-lxm* defines the class of uninflecting lexemes, and the symbol ! in MTR X ! Y means that the mother X is identical to the daughters except for the specification Y, in other words that the only difference between the daughter and the mother is that the daughter is of the supertype *lexeme*, while the mother is of the type *word*. The annotation ($\uparrow \text{infl-cxt}$) means that *zero-infl-cxt* is a subtype of the *infl-cxt* construction, (29), (Sag’s (57) Sag 2012: 108), which states “The mother of an inflectional construct is of type *word*; the daughters must be lexemes.”[Footnote omitted]

$$(29) \quad \text{Inflection construction in SBCG} \\ \text{infl-cxt: } \left[\begin{array}{l} \text{MTR } \text{word} \\ \text{DTRS } \text{list} : (\text{lexeme}) \end{array} \right]$$

The problems with the type shifting account, whether in Sign-Based Construction Grammar or in PFM, are the following. The principal problem is that it is clearly no more than an arbitrary device deployed solely to meet the parochial architectural needs of a particular formalization of the model. Second, it fails to capture the very clear intuition that an uninflecting lexeme is a lexeme that fails to inflect, not a lexeme with some strange ‘zero’ inflection. In particular, what feature specification is an uninflecting lexeme associated with? Do all uninflecting lexemes ‘realize’ the same ‘zero feature’? The counter-intuitiveness of the solution becomes obvious when we consider languages such as Vietnamese, which have no inflectional morphology at all. The type-shifting procedure would have to create a ‘trivially’ inflected form for every lexeme in the language, even though the language lacks any inflection. Finally, we have a question which is not addressed in Sag’s account, but which is central to this paper: what is the relationship between lexemes which belong to a non-inflecting class such as English prepositions (members of Sag’s *invariant-lxm* type, our uninflecting lexemes),

and uninflectable lexemes, whether lexically uninflectable, like KENGURU or contextually/constructionally uninflectable, like German predicative adjectives?

An alternative to a type-shifting account, sketched in Spencer (2019a,b) says that an uninflecting lexeme has literally no inflectional paradigm, whether Π_C or Π_F . It is not associated with any Π_C in any form, because it is not, by definition, associated with any morphologically expressed morphosyntactic property. A fortiori it is not associated with a Π_F . This leaves the question of how to specify the form of such a lexeme when it occurs in a syntactic construction, that is, how to specify lexical insertion for such a lexeme. Ideally, we should do this without appeal to the PFM equivalent of the type shifting operation (28).

In Spencer (2019a, 2019b; see also Spencer 2020) I proposed a Default Exponence Principle (DEP), which states informally that, by default, the form of any word undergoing lexical insertion (however this notion is characterized) is the lexical root, even in the case of a standardly inflecting lexeme. To see how that account was supposed to work let Σ be the set of features defining a content paradigm, and let T be the set of features defining the corresponding form paradigm. As things stand the Paradigm Linkage mapping will be undefined for cases where $\Sigma = T = \emptyset$. Spencer (2020) therefore assumes mapping (30), the Default Exponence Principle, as the default morphosyntactic expression of all lexemes defined over null feature sets.

(30) Default Exponence Principle (DEP) [to be rejected!]

$$\text{PF}(\langle Z, \emptyset \rangle) = Z = \text{STEM}_0(\mathcal{L})$$

The intended upshot is that the default realization of all lexemes is STEM_0 ($\equiv Z$, the lexical root). For inflecting lexemes the DEP (30) is overridden by the more specific *Corr* function. However, for an uninflecting lexeme DEP has the same effect as Sag's Zero Inflection Construction. This would mean that we would not require uninflectable lexemes to have a non-null inflectional paradigm. The uninflected lexemes whose form is defined by (30) have just a root form but no associated paradigm, content paradigm or form paradigm.

The idea behind this formulation is that the DEP is overridden by any genuinely inflected word form, in the sense that any lexeme which has a Π_F will have a definition of lexical form which is more specific than the DEP, and this will pre-empt the default. However, there is an important flaw in this way of viewing default root exponence. The difficulty relates to the apparently trivial and obvious aspect of the model mentioned above: what is the relationship between the lexical insertion of an inflected form and lexical insertion of an uninflected form? In the present case, the technical problem with our informal DEP is that it

is not possible to treat uninflected default root forms and inflected word forms as being in (Paninian) competition with each other. Therefore, it is not possible for a specific inflected form to override the root insertion default. Such an override would only be possible if the root insertion were formalized as a kind of zero inflection, of the sort we have just rejected.⁷

I therefore abandon the proposal based on the DEP. In order to account for uninflectability, however, we need to make completely explicit what we mean by lexical insertion. I shall sketch such an account, what I take to be essentially the notion of lexical insertion that is invariably presupposed in the literature, and then show how it serves as the basis of an account of uninflectability. My account will appeal crucially to the model of lexical representation and lexical relatedness offered in Spencer (2013).

The intuition we wish to capture is very simple: by default, lexemes are uninflecting and the phonological form which realizes a syntactic node, ν , is the sole phonological form associated with that lexeme, the lexical root. Where we have an inflecting lexeme the default root insertion operation has to be pre-empted by a more specific operation which inserts a more narrowly specified phonological form. That form is the Π_F correspondent of the most narrowly specified cell in the lexeme's Π_C corresponding to the featural content of syntactic node ν . I shall code the notion of lexical insertion in relatively informal terms, borrowing and adapting the notational conventions introduced in the Parallel Architecture model of Jackendoff & Audring (2020).

First, we assume that lexical insertion is coded as a co-indexing of a syntactic node, ν , with a morphophonological representation of a word form, ω . Thus, to say that the forms /ðə/, /blak/, /dog/ are inserted into the NP frame $[_{NP}[_D \text{ the}] [_A \text{ black}] [_N \text{ dog}]]$ is simply to provide the appropriate indices: $[_D \text{ the}]_1 [_A \text{ black}]_2 [_N \text{ dog}]_3 \Leftrightarrow \text{/ðə/}_1 \text{/blak/}_2 \text{/dog/}_3$.

Second, we should observe that such co-indexation is always feature-driven in the sense that the node ν is necessarily specified by at least a lexical category feature (or equivalent), and that the phonological form it is coindexed with has to be associated with a compatible feature specification. The leading idea is this. Let us denote the set of all the morphosyntactic, morpholexical interface features (morphosyntactic features) as $\mathcal{I}$. The set $\mathcal{I}$ will contain all the possible

⁷The appeal to the DEP also makes it more difficult to understand the relationship between uninflectedness and defective lexemes, since one of the entailments of DEP is that all licensed collections of σ -features for a lexeme correspond to some form, and this entailment is incompatible with the existence of defective lexemes. However, the question of defectiveness and uninflectability has to remain the subject of work in progress.

features which could govern morphosyntactic structure, including complementation/selection features and others, perhaps including sociolinguistic properties where these interact with morphosyntactic well-formedness. Lexical insertion is then a relation which takes a node in a syntactic structure and matches it with a word form or set of word forms which offer the best fit with the specification on that node. Where the syntactic construction is underspecified we may find that this partial specification or description is compatible with more than one form. For instance, in the syntactic context *The _____ sat on the mat* the unspecified position is compatible with a noun in either singular or plural form. However, for simplicity of exposition let us assume that the syntactic node is fully specified and that the semantics specifies a lexeme with a unique meaning, so that lexical insertion gives a unique output up to perfect synonymy, including semantically neutral overabundance.

(31) Lexical Insertion (LEXIN)

Given a syntactic node, ν , defined by some feature set $\mathcal{F} \subset \mathcal{I}$ and given a lexeme, lexemic index $\mathcal{L}$. Then $\mathcal{L}$ and ν can be coindexed iff there is some ω , a phonological word form of $\mathcal{L}$, which expresses/realizes the largest set $\mathcal{G} \subset \mathcal{I}$, of features such that $\mathcal{F} \subseteq \mathcal{G}$.

As formulated in (31) LEXIN will also license the lexical insertion of an uninflecting form such as *on* in *The cat sat _____ the mat*. In the case of a singular or plural noun the coindexation takes place between the syntactic node and the form correspondent of the appropriate cell in the lexeme's Π_C , which by Paradigm Linkage maps to the appropriate cell in the Π_F . In the case of an uninflecting preposition, I have argued that there is no Π_C , hence, no Π_F , so that the only form available for insertion will be the lexical root, realizing the purely syntactic features of the preposition, such as its lexico-syntactic category, P.⁸

The important point about the formulation of LEXIN is that it makes absolutely no (direct) reference to inflection. However, the set $\mathcal{I}$ clearly includes the set of (non-morphomic) inflectional features which define the content paradigms of lexemes, and which therefore serve as (part of) the interface between morphology and syntax. It is thus reasonable to say that the lexical insertion of an inflected word form is a special case compared to the insertion of an uninflecting word form. The featural characterization of an inflected word form will include as a

⁸In a complete formalization care will have to be taken to ensure that the morphophonological characterization of the inflected word forms in the realized paradigm, and the morphophonological characterization of the bare root form of an uninflecting lexeme are couched in the same morphophonological vocabulary, so that both types of representation will be "pronounceable" at the appropriate level of description.

bare minimum the categorial information associated with that lexeme (however such categories are defined) together with its own inflectional content.

We now have nearly all the machinery need to account for uninflectability. I will crucially appeal to the $\Pi_C \sim \Pi_F$ distinction, though, as far as I can tell, all that is required for the analysis to work is any equivalent distinction between syntactically relevant and syntactically invisible (morphomic) features, such as the morphological vs. syntactic feature distinction of Sadler & Spencer (2001) or the HEAD vs. INFL features distinction of HPSG-oriented treatments of morphology such as IBM (Crysmann & Bonami 2016, Crysmann 2020). In this respect, the problem of lexical and contextual uninflectability provides additional strong support for some such distinction.

It remains for us to ensure that the mechanisms by which the inflected forms of an inflecting lexeme are defined are essentially the same as the mechanism by which the root form of an uninflecting lexeme is defined. The model of lexical organization proposed in Spencer (2013) provides the necessary machinery. We assume that all lexemes are individuated by means of a unique lexemic index, LI, $\mathcal{L}$. For simplicity, in the case of concrete lexemes I will use the name of the lexeme in SMALL CAPITALS as its lexemic index: SHEEP, HIT etc. We further assume that a lexeme has to be furnished with at least three more representations or attributes: FORM, specifying as a minimum the unpredictable aspects of the lexeme's phonology and morphology, such as the shape of the root, any idiosyncratic stem forms or inflected forms, and so on; SYN, a representation of crucial aspects of the syntax of the lexeme, such as syntactico-lexical category membership, argument structure or complementation features and so on; SEM, a semantic/pragmatic representation. These representations can be thought of as the output of three functions over the LI, e.g. FORM(HIT), SYN(HIT), SEM(HIT). Consider an uninflecting lexeme such as the preposition UNDER. The morphophonological root form, /ʌndə/ (or whatever) is defined by FORM(UNDER). For an inflecting lexeme such as DOG we have a (maximally simple) Π_C , say, $\{\langle \text{DOG}, \text{SG} \rangle, \langle \text{DOG}, \text{PL} \rangle\}$. English morphology defines the *Corr* function in such a way that the Π_R of DOG is $\{\langle \text{dog}, \text{SG} \rangle, \langle \text{dogz}, \text{PL} \rangle\}$.

Our aim now is to provide a mechanism for the formal representation of an uninflecting lexeme such as UNDER and the formal representation of an inflecting lexeme such as DOG which allows both type of lexeme to be treated in the same way by the principles of lexical insertion. In effect, this amounts to uniting the FORM attribute for lexemes such as UNDER, and the *Corr* function as applied to inflected words, but without introducing a (covert) disjunction. We can achieve this by defining lexical insertion over an expanded lexical representation of inflected lexemes in which the content of the FORM attribute is generalized so as

to cover all possible form correspondents of a lexeme, including those defined initially in terms of a (non-trivial) Π_C . In other words, the expanded lexical entry of a lexeme incorporates that lexeme's realized paradigm, effectively coding the common notion of "full-listing" of inflected forms in lexical representations (Butterworth 1983).

This expanded lexical representation effectively formalizes the traditional notion that one of the purposes of the lexeme concept is to unify all the inflected forms of a lexeme as members of a single unit. In fact, such a generalization seems to be required even for uninflecting lexemes. For instance, some lexemes exhibit allomorphy which is determined in purely morphophonological terms independently of morphosyntactic context. The English indefinite article, *a~an* is a case in point. The FORM attribute applied to *A*, the LI of the indefinite article, therefore yields two outputs, $\{/a/, /an/\}$, perhaps together with some morphophonological featural annotation specifying which contexts each form occurs in.⁹ Other cases of this sort might be found with uninflecting lexemes that have full form vs clitic alternants, high vs low register variants, and so on, or more exotic alternations of the kind illustrated by English *in my stead ~ instead of me*. Such alternations are not inflectional in character, but they do show that defining the root form of an uninflecting lexeme can be non-trivial. Against this background we now consider (non-defective) inflecting lexemes.

Consider the 2nd declension Latin noun *DOMINUS*, illustrated earlier in Table 9. Let's assume that its root form is */domin/*. We now appeal to the architecture of lexical representation proposed in Spencer (2013: 184–89), in which the FORM attribute includes the lexeme's morphological signature, *MORSIG*, a subattribute of FORM, which specifies the morphological class (*MORCLASS*) to which the lexeme belongs for the purposes of the inflectional morphology. For instance, the lexical representation for the Latin *DOMINUS* will have a FORM attribute roughly as in (32).

$$(32) \left[\text{FORM} \begin{bmatrix} \text{MORSIG} & [\text{MORCLASS } \textit{declension2}] \\ \text{STEM}_0 & \textit{domin} \end{bmatrix} \right]$$

The attribute FORM for *DOMINUS* will specify this stem (directly) and also all the inflected forms of Table 9 (indirectly, via the rules of Latin morphology). This gives us the schematized lexical representation shown in (33), where $\langle \dots \rangle$ stands for the representations of the cells of the rest of the paradigm.

⁹This may mean that FORM is a relation rather than a function, *sensu stricto*. However, this depends on exactly how such allomorphy is implemented, and so I will continue to speak of functions here.

$$(33) \left[\text{FORM} \left[\begin{array}{l} \text{MORSIG} [\text{MORCLASS } \textit{declension2}] \\ \text{STEM}_0 \textit{ domin} \\ \text{NOM,SG } \textit{dominus} \\ \text{NOM,PL } \textit{domini:} \\ \langle \dots \rangle \end{array} \right] \right]$$

Similarly, the expanded FORM attribute for the German adjective KLEIN will be roughly that of (34).

$$(34) \left[\text{FORM} \left[\begin{array}{l} \text{MORSIG} [\text{MORCLASS } \textit{adjective}] \\ \text{STEM}_0 \\ \text{MASCULINE NOM.SG} \\ \text{NEUTER NOM.SG} \\ \langle \dots \rangle \end{array} \right] \begin{array}{l} \textit{klein} \\ \textit{kleiner} \\ \textit{kleines} \end{array} \right]$$

In both the Latin and the German examples the STEM_0 form is the basis for the inflectional rules of morphology. The crucial difference is that in the case of KLEIN, the German morphosyntax includes rules which make direct appeal to the STEM_0 form, whereas this is not true for Latin, even if that STEM_0 form happens to have the morphophonology of a pronounceable word.¹⁰

Now consider an uninflectable lexeme such as Russian KENGURU. I assume that as a lexical property its *Corr* function fails to define a mapping to Π_F . This means that none of the cells in the lexeme's Π_C has a form correspondent. We code this fact by stipulating that KENGURU belongs to an inflectional class [MORCLASS:uninflectable]. Lexemes of this class are associated with a property mapping which states that the feature set for the form/realized paradigm of the lexeme is undefined: $pm(\sigma) = \text{undefined}$. Effectively, this means that there will be no expanded lexical entry for KENGURU, so that its lexical representation will remain that shown in (35).

$$(35) \left[\text{FORM} \left[\begin{array}{l} \text{MORSIG} [\text{MORCLASS } \textit{uninflectable}] \\ \text{STEM}_0 \textit{ kenguru} \end{array} \right] \right]$$

However, our definition of lexical insertion makes no direct mention of a form correspondent to an inflectional cell, or indeed to any kind of paradigm, whether

¹⁰This account opens up the possibility that a lexeme might have two STEM forms, one the basis for morphology, whether free or bound, the other specifically applicable only to certain syntactic constructions. The special combining forms found in some of the compounds of German (Nübling & Szczepaniak 2013) and other languages might be a candidate.

Π_C , Π_F , or Π_R . LEXIN just asks for a phonological word associated with the features on the syntactic node. In the absence of the (generally very specific) form-feature pairings defined by the Π_F , Π_R , all we have is the lexical root (or what I have labelled ‘STEM₀’). It is therefore this root that undergoes the coindexation which instantiates lexical insertion in our informal characterization. It is in this sense that the root of KENGURU is a default form for inflection. The effect is to define a lexeme which behaves morphologically like an uninflecting lexeme, but which can undergo lexical insertion just as though it were an inflecting lexeme. The case of Russian female referent foreign names such as (*Hillary*) *Clinton* can be dealt with in the same way that we handle the *kenguru*-nouns, provided the grammatical representation of proper names has access to the referent’s sex/gender, and provided we allow the female referent form of the name to constitute a distinct lexeme from the male referent form. Note that this only applies at the level of individual name lexemes; it does not apply to the class of non-native female names as a whole, or to the entire NP containing that name. A woman with a Russian (-sounding) given name, such as *Melania Trump* or *Marina Warner* will have that given name inflected but not the surname: *kniga Melani-i Trump/Marin-y Vorner* ‘Melania Trump’s/Marina Warner’s book’.

In the case of contextual uninflectability the main descriptive burden generally has to be placed on the syntax. Thus, the obvious way of handling the contextually uninflecting German predicative adjective is to stipulate that in predicative use (however defined) LEXIN targets that value of the adjective’s FORM attribute which has just categorial features and no others, namely, the STEM₀. This will mean that none of the features in the adjective’s Π_C will be applicable, so that in predication position the adjective will behave exactly like an uninflecting (NB *not* uninflectable) lexeme. Again, since LEXIN is effectively defaulting here to the lexical root, the grammar will not have to pretend that *klein* in (7b) (repeated here in adapted form) bears any (spurious) inflectional features.

- (7b) Das Känguru ist **klein**.
 the kangaroo is little(UNINFL)
 ‘The kangaroo is little.’

This makes the case of predicative adjectives identical to that of other complement predicates to copular verbs in German. For instance, in (36) we see an instance of a NP functioning as the complement to the copular.

- (36) a. Das Känguru ist mein Freund.
 the kangaroo.SG is my.M.NOM.SG friend(M).NOM.SG
 ‘The kangaroo is my friend.’

- b. Die Kängurus sind unsere Freunde.
the kangaroo.PL are our.NOM/ACC.PL friend.NON-DAT.PL
'The kangaroos are our friends.'

The complement NPs *mein Freund* 'my friend' and *unsere Freunde* 'our friends' are in the nominative singular/plural form (unambiguously so in the case of the singular). Here we have to say that the syntax demands a nominative case marked predicative NP complement. This is not a given. In Polish, for instance, the NP complement to any copular verb will normally be in the instrumental case, while in Russian the choice of case depends on lexical and semantic factors. Adverbial phrases, including prepositional phrases, can also serve as complements to the copular verb, as seen in example (37) from German.

- (37) Das Känguru ist hier/hinter dem Baum.
the kangaroo is here/behind the tree
'The kangaroo is here/behind the tree.'

In such constructions the adverbials *hier*, *hinter dem Baum*, do not realize any morphosyntactic property at all.

We have now largely achieved one of our principal aims. In both Type 1 and Type 3 total uninflectability LEXIN targets an uninflected stem form, either because the syntax demands it (contextual/constructional uninflectability) or because the lexeme lacks any other forms (lexical uninflectability).

Turning now to partial uninflectability, consider first the Polish *muzeum*-words. Recall that lexemes of this class, MUZEUM, STUDIO and so on, typically retain, it seems, their latinate morphological structure: *muze-um*, *studi-o*. This can be seen from the plural inflected forms: *muze-a*, *muze-ów*, ..., *studi-a*, *studi-ów*, These are based on a bound stem, *muze-*, *studi-*, which also appears in derived forms: *muze-al-n-y* 'pertaining to museums'. In effect, such lexemes exhibit heteroclis. The plural part of the paradigm inflects according to a relatively standard inflectional class (though with an unusual genitive plural form), based on a bound stem. The singular part of the paradigm is uninflectable, and is based on what historically is generally the nominative singular ("citation form") of the original Latin word, i.e. that part of the paradigm is governed by the feature MOR-CLASS:uninflectable. Thus, the singular and plural paradigms belong to distinct inflectional classes. More specifically, following Stump (2016: 189–91) such a lexeme is associated with two distinct (though in this case, phonologically related) basic stems, each of which belongs to a distinct inflectional class. The singular subparadigm is defined over the invariant stem form *muzeum*, and this stem is

marked as being uninflectable. This means that it is not associated with any Π_F . In the plural, however, the stem, *muze-* inflects according to the specific inflectional class it belongs to.

Consider again the phenomenon of ‘sporadic agreement’ (Fedden 2019). There are two important differences between sporadic agreement as seen in, say, the Nakh-Dagestanian languages and the partial uninflectability seen in cases such as the Polish *muzeum*-class. The *muzeum*-class words are very much the exception in the Polish lexicon, and for half their paradigm they are completely uninflectable. However, in the case of sporadic agreement it is generally the verbs showing agreement morphology that are in the minority, and those verbs that fail to show agreement nonetheless inflect normally for other inflectional categories. Thus, there is a sense in which sporadic agreement is a species of overdifferentiation, though on such a scale that it has to be coded explicitly in the grammar and not just ignored as an aberration. But this simply means that the agreeing lexemes are given a richer *morpholexical signature* than other lexemes, that is, they are associated with a richer set of inflectional features for which they are obliged to inflect (here, agreement features). As far as I can tell, the sporadic agreement cases do not therefore pose any special problems compared with other types of partial uninflectability: only a part of the verb lexicon bears specification for the agreement features.

Now consider again the Polish *widmo*-construction. The nominative singular form *widmo* (and likewise, the form *muzeum*), is, of course, traditionally regarded as the citation form of the lexeme WIDMO. It is presumably for this reason that such forms are chosen to serve as uninflecting variants in languages such as Polish or Latin, whose inflectional stems are generally bound morphemes. In such languages, citation forms seem the natural choice of default form with respect to lexical insertion. However, from a formal point of view this seems to entail that lexical insertion can in part be defined over the lexeme’s morpholexical signature, i.e. the set of inflectional features for which it must inflect. Specifically, it appears to require that default application of LEXIN can appeal somehow to those feature sets which function as defaults. It is not obvious how such an effect can be achieved formally, however.

A somewhat similar situation is found when we see the use of a citation form in titles of literary works and similar.¹¹ As a consequence, in certain contexts such titles and similar types of name (e.g. names of journals or newspapers), appear to be uninflecting in languages where we could otherwise expect them to inflect. In Russian, for instance, the title *Vojna i mir*, ‘War and peace’, inflects as a normal

¹¹I am grateful to Grev Corbett for urging me to consider such constructions.

conjoined noun phrase, as in (38), with both nouns appearing in the expected genitive case.

- (38) poslednee izdanie “Vojn-y i mir-a”
latest edition War-GEN.SG and peace-GEN.SG
‘the latest edition of “War and Peace”’

However, somewhat more natural, perhaps, is a construction in which the title is in apposition to a generic noun such as ROMAN ‘novel’, as in (39).

- (39) poslednee izdanie roman-a “Vojn-a i mir”
latest edition novel-GEN.SG War-NOM.SG and peace.NOM.SG
‘the latest edition of the novel “War and Peace”’

In (39) it is the noun ‘novel’ which receives genitive case, while both nouns in the title itself remain in the nominative singular citation form.

The appositional construction illustrated in (39) is reminiscent of compound nouns in a number of languages with a phrasal non-head, such as *his why-does-it-always-have-to-happen-to-me attitude*.¹² One class of such phrasal compounds is generally taken to be quotative: the phrasal non-head expression is not used as such, but rather quoted. If we take the title *Vojna i mir* in (39) to be the citation of the title then it is less mysterious to see it appearing in the citation form. It might even be possible that citation accounts for the use of the default nominative singular in the Polish *widmo*-constructions, though further speculation would take us too far afield.

Finally, we turn to the cases of partial constructional/contextual uninflectability, as exemplified by the Hungarian agreeing infinitive and the Latvian genitive compounds. The Latvian genitive compounds exhibit a structure which is in many ways comparable to the compounds in which the non-head of the compound has to appear in a special “combining form”, as found in Sanskrit, Greek, German and other languages. For instance, German has a large number of compounds of the form *Liebeslied* ‘love song’, formed by combining the nouns *Liebe* ‘love’ and *Lied* ‘song’ with a meaningless intermorph *-s-*: *Liebe-s-lied* (Nübling & Szczepaniak 2013), and many Russian compounds take the form of the example (23) *klin-o-obraznyj* ‘wedge-shaped’ cited above. The crucial difference is that the Latvian ‘combining form’ is always identical to a genitive case form of the non-head noun (singular or plural). This means that the grammar has to select

¹²See the contributions to Trips & Kornfilt (2017) for discussion of phrasal compounding and specifically Pafel (2017) for interesting typological observations.

a specific cell in the lexeme's paradigm, as opposed to a particular stem form or the root form.

The Hungarian infinitive construction is essentially homologous to the spontaneous agreement described for Nakh-Dagestanian languages and others, with the difference that the construction applies to all verbs and not just to a (sizeable) minority. The solutions proposed for spontaneous agreement will therefore presumably carry over to the Hungarian case, but without the lexical restrictions needed for lexically-governed spontaneous agreement.

5 Conclusions and queries

This paper has had two principal goals.

First, I have extended the typology of uninflectability, arguing for the four-way typology shown in Table 2. This establishes two parameters of uninflectability. A word can be either totally or partially uninflectable. A word can be uninflectable either as an inherent lexical property or the uninflectability can be induced by the syntactico-semantic context (contextual uninflectability) or actually introduced as part of a grammatical construction (constructional uninflectability). The literature attests only three of the four possibilities, but additional evidence from Hungarian and Latvian suggest that the full typology is viable.

As part of this conceptual endeavour I have asked just what inflectional features it is that a lexeme can be uninflectable for. The answer seems to be just the core inflectional features that are associated in that language with (more or less canonical) synthetic morphology.

Second, I have asked how uninflectability can be accommodated in a formal grammar. Taking as my point of departure the PFM2 model of Stump (2016) I have argued that we can treat uninflectability as a total or partial lack of a form/realized paradigm in a lexeme that nonetheless has a fully-fledged content paradigm. The fact that the uninflectable lexeme is associated with a content paradigm distinguishes the uninflectable lexemes from the common-or-garden uninflecting classes of lexemes, such as English prepositions. By defining lexical representations in the right way, however, we can make the principles of lexical insertion (the syntactic licensing of a concrete form of a lexeme) apply in the same way to uninflecting lexemes and uninflectable lexemes. By defining the FORM attribute of a lexeme as the output of inflectional morphology it is then also possible to account for the lexical insertion of ordinarily inflected word forms using the same machinery.

A number of questions arise from this study.

I have suggested, on the basis of admittedly limited evidence, that uninflectability only applies in the case of thorough-going synthetic inflection. To what extent is this actually true? And to the extent that it is true, does it provide a test bed for identifying the notion of *core synthetic inflection* in a given language? If so, this might have implications for the notion of paradigmatic organization and for the putative distinction between inflection and derivation.

Uninflectability is not an all-or-nothing, binary phenomenon. It remains to be seen, therefore, whether the account sketched here can accommodate all gradient cases of uninflectability in a convincing fashion.

The PFM2 model presupposes a very specific view of inflectional morphology, and the level of form paradigm, in a sense, is an artefact of that model. Other closely related models do not appeal to such a level, for instance, Network Morphology (Brown & Hippisley 2012) or the morph-based HPSG-oriented implementation of PFM2 in Information-based Morphology (Crysmann & Bonami 2016). Moreover, the notion of form paradigm itself raises conceptual questions: if the feature ensemble defining the content paradigm has to be interpreted by syntax/semantics then those features cannot, by definition, be morphomic (in the sense of Aronoff 1994 and many subsequent references, including Stump 2016). But if the form paradigm represents a level that is ‘informationally encapsulated’ from syntax/semantics and just defines formal, morphological relations, then it is odd to see that the features which define it in the canonical cases of linkage are precisely the non-morphomic content paradigm features (cf Boyé & Schalchli 2019, Spencer 2021). Future research should therefore be devoted to generalizing the account offered here so as to make it compatible with any model of inferential-realizational inflectional morphology.

Finally, we have seen that it is important to distinguish uninflectability from defectivity, but does the current proposed analysis of uninflectability have any (possibly unwelcome) implications for defectivity? A defective lexeme is one which lacks one or other forms from its paradigm, so that there is no way of expressing that particular cell’s feature content for that lexeme. For instance, the Russian noun МЕЧТА ‘dream’ cannot be used in the genitive plural form, even though all Russian speakers know what that form ‘ought’ to be, namely, *[!]mečt (where the annotation *[!] is intended to mean ‘form which is expected, but which is unacceptable’). Stump (2010), examining the interaction of defectiveness with syncretism, suggests that defectiveness might be a property either of content paradigms, or of form paradigms. My approach to uninflectability suggests that it is better to treat defectivity as a property of the lexeme’s content paradigm, as the safest way of ensuring that LEXIN has no member of the form paradigm/FORM

attribute to fall back on. In other words, a defective lexeme is one with a defective morpholexical signature, which simply denies the existence of certain cells in the content paradigm, hence, a priori in the form/realized paradigm.

Abbreviations

ACC	accusative	LOC	locative
<i>Corr</i>	Correspondence (function)	LVC	light verb construction
DAT	dative	M	masculine
DEP	Default Exponence Principle	MWE	multiword expression
F	feminine	N	neuter
GEN	genitive	NOM	nominative
HPSG	Head-driven Phrase Structure Grammar	PF	Paradigm Function
IbM	Information-based Morphology	PFM	Paradigm Function Morphology
IFD	Identity Function Default	PL	plural
INF	infinitive	<i>pm</i>	property mapping
INS	instrumental	PX	possessor
LFG	Lexical Functional Grammar	SBCG	Sign-Based Construction Grammar
LI	Lexemic Index	SG	singular
		TAM	tense-aspect-mood
		UNINFL	uninflectable
		VOC	vocative

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